Real-Time PPE Detection Using Computer Vision

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- Computer Vision
- Tier 2 Project

The Problem

- Many workplaces like warehouses and construction sites require workers to wear personal protective equipment (PPE).
- Supervisors cannot constantly monitor every worker.
- Workers sometimes forget helmets or safety vests, increasing the risk of injuries and safety violations.

Project Solution

- Build a computer vision system that detects PPE equipment in images or video.

- Workflow:
- Camera Image → Object Detection Model → Detect Helmet & Vest → Alert if PPE is missing

Technical Approach

- Computer Vision Technique: Object Detection
- Model: YOLOv8
- Framework: Python + PyTorch + Ultralytics

- Reason: YOLOv8 is fast and accurate for detecting objects in real-time.

Data Plan

- Dataset Source: SH17 Dataset for PPE Detection
- Dataset Size: 8099 Images

- Labels:
- - Person
- - Helmet
- - Safety Vest

System Pipeline

- Input Image or Video Frame
- ↓
- YOLOv8 Object Detection Model
- ↓
- Detect Person, Helmet, and Vest
- ↓
- Display Bounding Boxes and Alerts

Success Metrics

- Primary Metric: Mean Average Precision (mAP)
- Target: 85–90% detection accuracy

- Secondary Metrics:
- - Inference time per image
- - Real-time detection capability

Week-by-Week Plan

- Week 10: Collect dataset and set up environment
- Week 11: Train initial model
- Week 12: Improve accuracy and tune parameters
- Week 13: Test and evaluate model
- Week 14: Build demo
- Week 15: Final presentation

Challenges & Backup Plans

- Challenge: Small dataset → Use data augmentation or more datasets
- Challenge: Low accuracy → Try larger YOLO model
- Challenge: Training slow → Use Google Colab GPU

Resources Needed

- Compute: Google Colab GPU
- Tools: Python, PyTorch, Ultralytics YOLOv8
- Estimated Cost: $0
- Datasets and tools are publicly available.